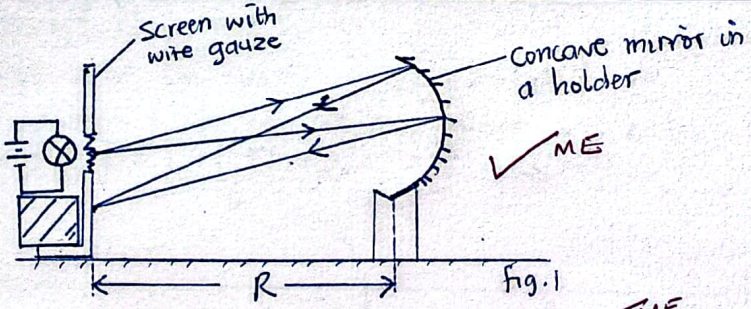
[redacted] GUIDE FOR PHYSICS P510/2 – 2026

ITEM ONE

CODE	DESCRIPTION FOR THE CODE	EXPECTED RESPONSE	COMPUTATION	NO OF KEY POINTS	LEVEL
PE (Planning for the experiment)	Creates the; (i) aim of the experiment - quantity to determine - relationship between the variables - purpose (ii) Hypothesis - statement of the hypothesis - realistic value or range (iii) Variables - identifies the three variables with their examples.	Aim: An experiment to determine the focal length of the concave mirror by varying the object distance, u, and recording the image distance, V so as to ascertain the suitability of the reflector dish in replacing the stolen one. ✓PE Hypothesis: The focal length of the dish is in the range of 80cm to 120cm. ✓PE Variables. Independent: Object distance, u ✓PE Dependent: Image distance, V ✓PE Controlled/Fixed: Intensity of the light produced by the bulb ✓PE	2/3 x 5 1/3 x 5 1 0	3 to 5 2 1 0	3 2 1 0
ME (Method of the experiment)	(i) Drawing a well labelled and workable diagram for each method. (ii) Writing relevant and coherent steps in past tense (iii) appropriate use of apparatus or materials in the steps (iv) use of appropriate quantities	Method I  Fig. 1 ✓ME a) The apparatus were set up as shown in the figure 1 above ✓ME b) The position of the mirror was adjusted by moving it to and fro until a sharp ✓ME clear image of the wire gauze was formed besides it c) The distance, R, was measured ✓ME and recorded using a metre rule ✓ME	3/4 x 18 1/2 x 18 1/4 x 18 0	13 to 18 9 to 12 4 to 8 1 to 3 0	4 3 2 1 0

(v) identification of errors and precautions.

d) The focal length, f_1, was calculated from the expression $f_1 = \frac{R}{2}$ ✓ME

Method II

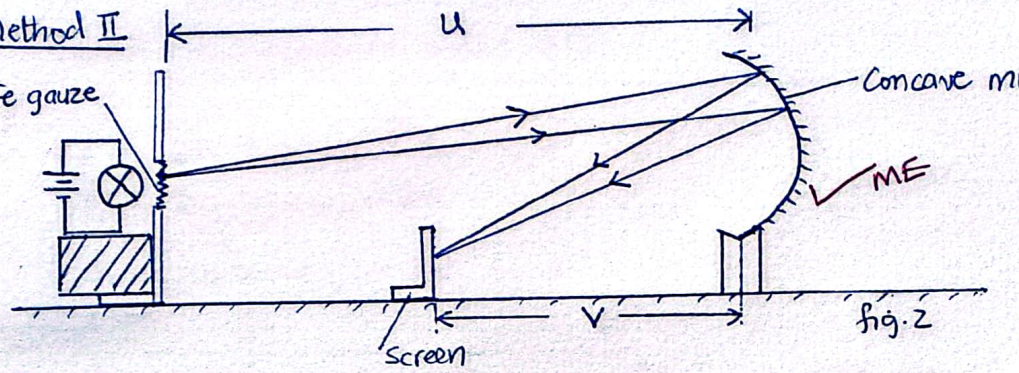


fig. 2

a) The apparatus were set up as shown in figure 2 above. ✓ME

b) With u = 25.0cm, the position of the screen was adjusted by moving it to and fro until a sharp clear image of the wire gauze was formed on it. ✓ME

c) The distance, V, was measured using a metre rule and recorded. ✓ME

d) Steps (b) and (c) were repeated for values of u = 30.0, 35.0, 40.0 45.0 and 50.0cm ✓ME

e) Results were recorded in a table including the values of $\frac{u}{v}$ ✓ME

f) A graph of u against $\frac{u}{v}$ was plotted. ✓ME

g) The intercept, c on the u-axis was read and recorded. ✓ME

h) The slope, s of the graph was calculated. ✓ME

(i) The focal length f of the concave mirror was calculated from the expression $f = \frac{f_1 + c + s}{3}$ ✓ME

Errors and Precautions.

Error	Precaution
i) Error due to parallax while taking readings from the metre rule ✓ME	- Positioning the eye at 90° to the scale of the metre rule when reading values
(ii) Too much light in the room ✓ME	- Regulating light in the room by closing the windows and switching off the bulbs.

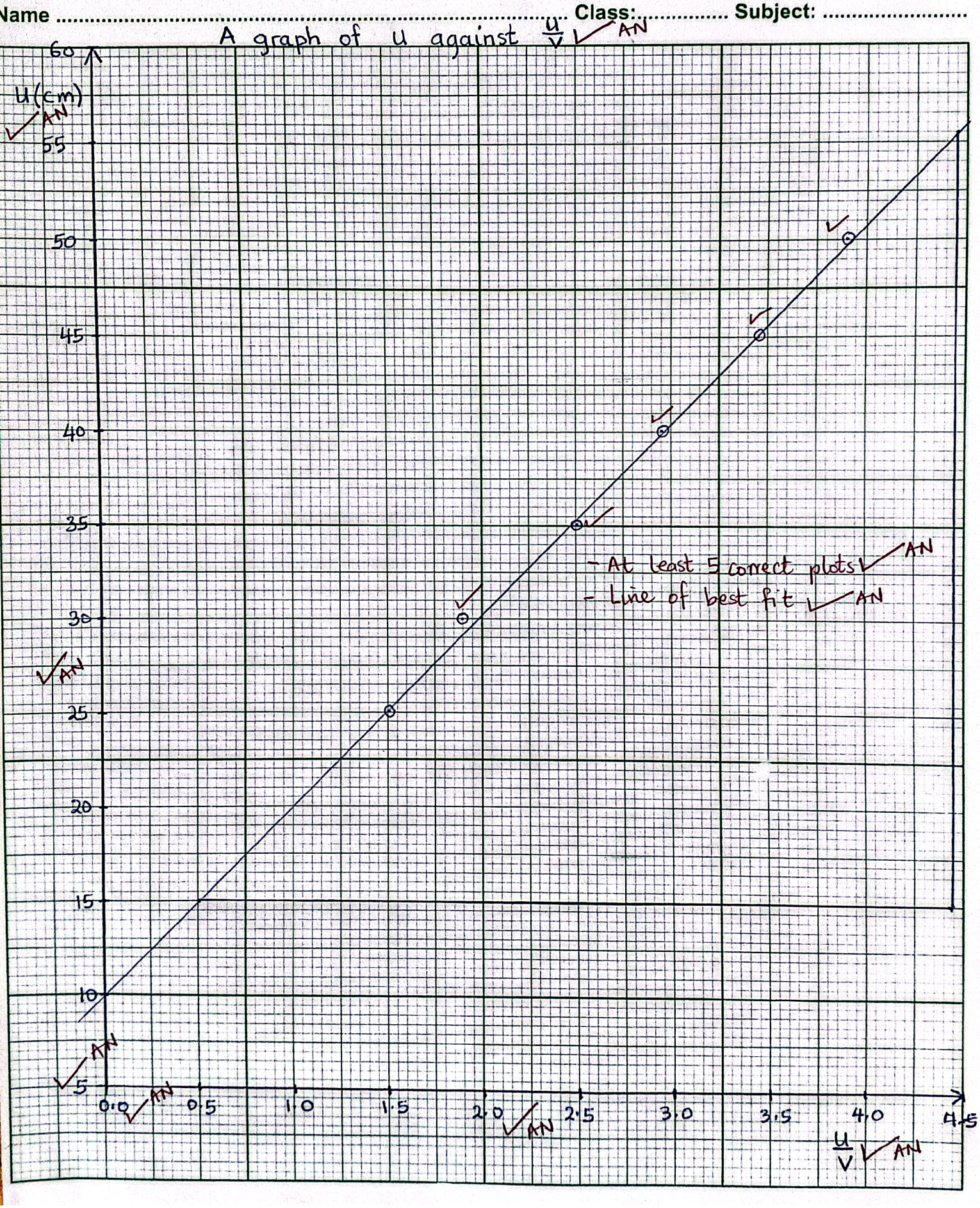
		NB- Consider any appropriate errors and their correct precautions given to this particular experiment. - Each instrument is scored once			
PD (Presentation of data)	(i) Recording, R to 1dp in cm (ii) Recording data in an organised format eg a table (iii) Presenting at least 5 data sets of independent and their corresponding dependent variables (iv) Writing quantities with their correct units in brackets where units exist (v) Recording the dependent variables accurately to 1dp (vi) The trend of the variables followed (vii) Correct data manipulation based on significant figures and decimal place values per column.	Method I $R = 20.4\,cm$ ✓PD $f_1 = \frac{20.4\,cm}{2}$ ✓AN $f_1 = 10.2cm$ ✓AN Method II (table below) - For five complete set ✓PD - all quantities with their correct ✓PD units in brackets ie u(cm), v(cm) and $\frac{u}{v}$ - recording v values to 1dp ✓PD - trend (if u values are increasing then v values should reduce and vice versa) ✓PD - calculation of $\frac{u}{v}$ ✓PD NOTE: In case the learner uses uv, (u+v) or $\frac{1}{u}$ and $\frac{1}{v}$, follow accordingly	$\frac{3}{4} \times 6$ $\frac{1}{2} \times 6$ $\frac{1}{4} \times 6$ 0	4 to 7 3 2 1 0	4 3 2 1 0

u(cm)	v(cm)	$\frac{u}{v}$
25.0	16.7	1.50
30.0	15.7	1.91
35.0	14.0	2.50
40.0	13.5	2.96
45.0	13.0	3.46
50.0	12.8	3.91

AN (Data analysis)	(i) Substituting in the formula $f_1 = \frac{R}{2}$ (ii) Calculation of the value for f_1 with the unit. (iii) Correct title of the graph	Intercept $C = 10.0cm$ ✓AN Slope $S = \frac{\Delta u}{\Delta \frac{u}{v}}$ ✓AN $S = \frac{55.5 - 15.0}{4.45 - 0.50}$ ✓AN ✓AN ✓AN ✓AN	$\frac{2}{3} \times 23$ $\frac{1}{3} \times 23$ 0	14 to 23 7 to 13 1-6 0	3 2 1 0

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	(iv) drawing and labelling axes correctly with units in brackets (v) uniform scale for each axis (vi) starting values for each axis (vii) at least 5 correctly plotted values (viii) line of best fit (ix) obtaining the slope (x) obtaining the intercept (xi) calculating the average of fs	$s = \frac{40.5}{3.95}$ ✓AN (rule of division) $s = 10.3\text{cm}$ ✓AN $f = \frac{f_1 + c + s}{3}$ $f = \frac{10.2 + 10.0 + 10.3}{3}$ ✓AN $f = \frac{30.5}{3}$ ✓AN (rule of addition) $f = 10.2\text{cm}$ ✓AN The focal length of the dish = $10 \times 10.2\text{cm}$ ✓AN $= 102\text{cm}$ ✓AN			
CR (Conclusion and recommendation)	(i) relates the obtained value of f to what is being investigated correctly. (ii) gives an appropriate advice or recommendation based on the value obtained.	The focal length of the dish is 102cm ✓CR which is within the range of 80cm to 120cm, therefore, the hypothesis is valid ✓CR. Hence, it can be used to replace the stolen one perfectly. ✓CR	½ x 3	2 to 3	2
			1	1	1
			0	0	0
		Note: No total is required. For the purpose of the mock, the total for this item is 16 ie $\frac{x}{16}$			

ITEM TWO

CODE	DESCRIPTION FOR THE CODE	EXPECTED RESPONSE	COMPUTATION	No OF KEY POINTS	LEVEL
PD (Presentation of data)	(i) Recording values of V and I to the required dps of the instrument for $d_1 = 30.0cm$ (ii) Recording values of V and I to the required dps of the instrument for $d_2 = 60.0cm$ (iii) Recording data in a table (iv) presenting at least 5 sets of independent and their corresponding dependent variables (v) Writing quantities with their correct units in brackets ie x(m), d_1(cm), d_2(cm) $\frac{d_1}{d_2}$ and R(Ω) (vi) Recording the dependent variables accurately with the correct dps of the instrument within the accepted range and the trend correctly followed (vii) Correct data manipulation based on significant figures for columns of $\frac{d_1}{d_2}$ and R	**Method I** (PD, PD, PD, PD)	$\frac{3}{4} \times 13$	10 to 13	4
			$\frac{1}{2} \times 13$	6 to 9	3
			$\frac{1}{4} \times 13$	3 to 5	2
				1 to 2	1
			0	0	0

d(m)	V(V)	I(A)
0.300	1.35	0.52
0.600	1.50	0.42

$$\mu_1 = \frac{1}{2}\left(\frac{V_1}{I_1 d_1} + \frac{V_2}{I_2 d_2}\right)$$

$$\mu_1 = \frac{1}{2}\left(\frac{1.35}{0.52 \times 0.300} + \frac{1.50}{0.42 \times 0.600}\right)$$ AN (substitution)

$$\mu_1 = \frac{1}{2}(8.7 + 6.0)$$ AN (rule of multiplication and division)

$$\mu_1 = \frac{1}{2}(14.7)$$ AN (rule of addition)

$$\mu_1 = 7.35 \Omega m^{-1}$$ AN

Method II

x(m)	d_1(cm)	d_2(cm)	$\frac{d_1}{d_2}$	R(Ω)
0.800	33.0	67.0	0.493	4.93
0.700	30.6	69.4	0.441	4.41
0.600	25.7	74.3	0.346	3.46
0.500	23.8	76.2	0.312	3.12
0.400	19.0	81.0	0.235	2.35
0.300	15.5	84.5	0.183	1.83

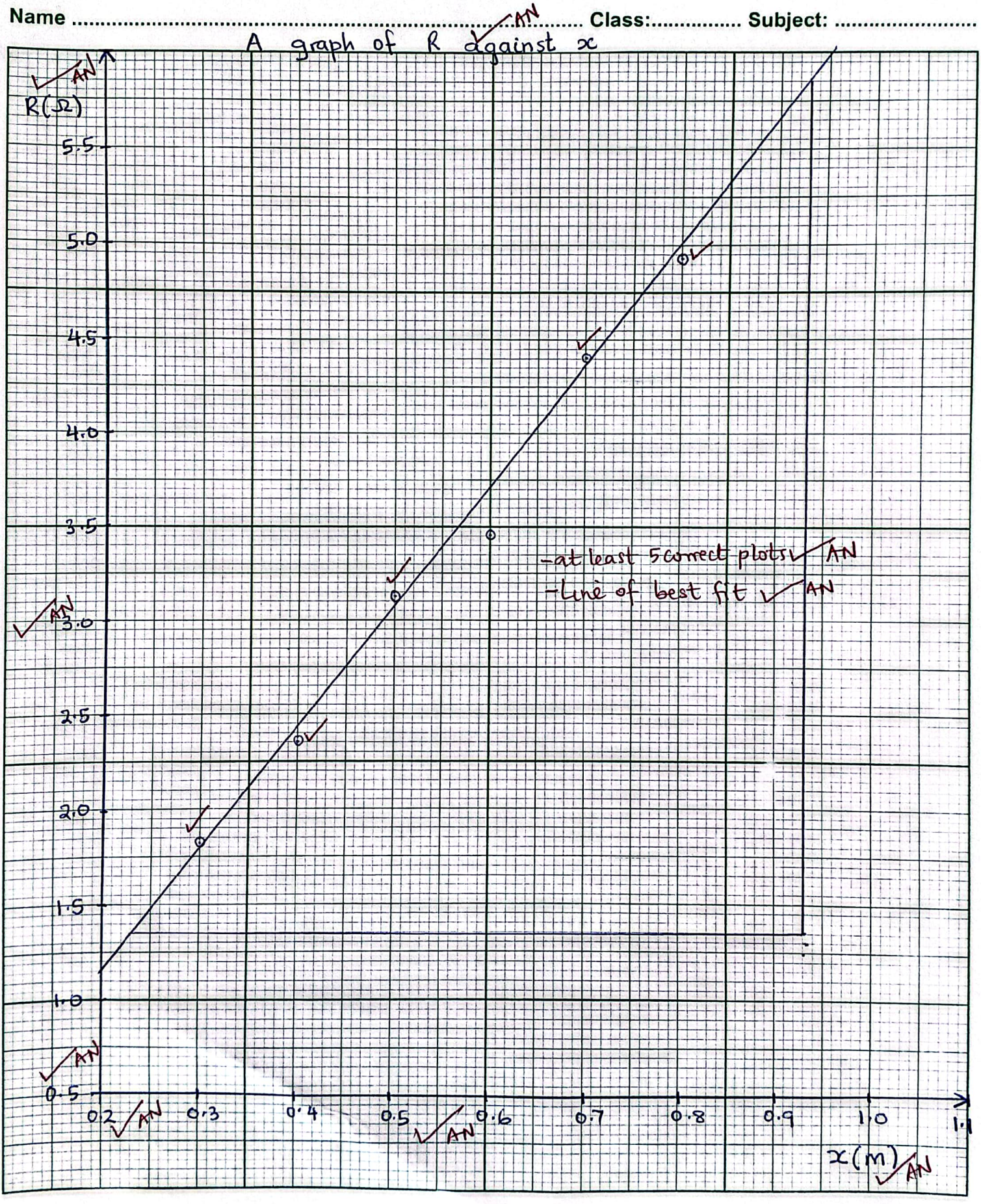
PD (labelling columns)

PD (5 sets of values) PD (d_1 values) PD (d_2 values) PD (calculation of $\frac{d_1}{d_2}$) PD (calculation of R)

PD trend (d_1 reducing while d_2 increasing)

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Errors	How it affects.
(i) Error due to parallax in taking readings from metre rule PD	- it lowers or raises the resistance per metre of the wire
(ii) Heating of the wires PD	- increases resistance per metre of the wire
(iii) Loose connection of the wire	- increases resistance per metre of the wire
(iv) Insensitivity of voltmeter and ammeter	- increases inaccuracy of values reducing or increasing resistance per metre

any 2

AN (Data analysis)

(i) calculation of μ_1
- substitution
- correct arithmetic and unit

(ii) Correct title without units
(iii) drawing and labelling of axes with correct units in brackets
(iv) uniform scale for each axis
(v) starting value for each axis
(vi) At least 5 correctly plotted values
(vii) line of best fit.
(viii) obtaining the slope or intercept
(ix) determining μ, correctly based on significant figures

Slope, $\mu_2 = \frac{\Delta R}{\Delta x}$ AN

$\mu_2 = \frac{5.85 - 1.35}{0.93 - 0.23}$ AN AN AN

$\mu_2 = \frac{4.50}{0.70}$ AN

$\mu_2 = 6.4\,\Omega m^{-1}$ AN

$\mu = \frac{\mu_1 + \mu_2}{2}$

$\mu = \frac{7.35 + 6.4}{2}$ AN

$\mu = \frac{13.8}{2}$

$\mu = 6.90\,\Omega m^{-1}$ AN

$\frac{3}{3} \times 21$	14 to 21	3
$\frac{1}{3} \times 21$	7 to 13	2
	1 to 6	1
0	0	0

CR (Conclusion and recommendation)

(i) relating the resistance per metre obtained with what is being investigated correctly.
(ii) giving an appropriate advice or recommendation based on value(s) obtained

The resistance per metre of the wire is $6.90\,\Omega m^{-1}$ CR which is above the required CR one for the examination. Therefore, the DOS should get another wire suitable for the Physics practical exam. CR

NOTE: Total not necessary but for purposes of calculation of percentage, it is out of 9.

$\frac{1}{2} \times 3$	2 to 3	2
	1	1
0	0	0

GRAND TOTAL FOR THE PAPER = 25